

SummarIQ: Equation-Aware Summarization of Scientific Papers

AI-Powered Summaries Using Transformer Models

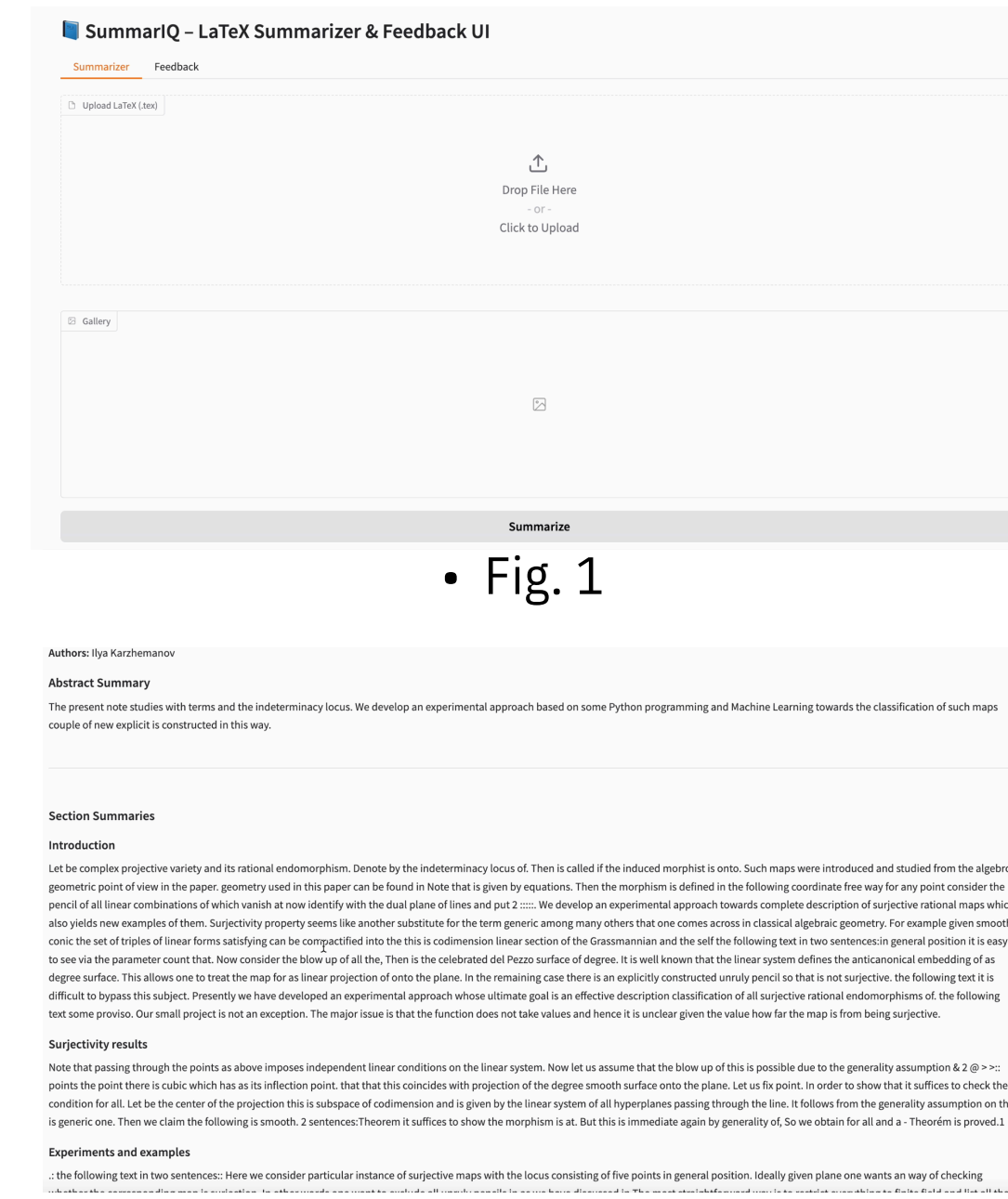
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Why SummarIQ?

- Reading LaTeX research papers is slow and difficult due to complex structures, dense **mathematical expressions** and technical formatting.
- Existing summarisation tools focus mainly on plain text and fail to **preserve equations, section hierarchy, and scientific notation**.
- Researchers, students, and engineers need **quick, accurate summaries** that still retain the essential mathematical context.
- Modern transformer models like **T5** and **BART** are powerful, but they require careful preprocessing to work effectively on LaTeX documents.
- SummarIQ** bridges this gap by providing an **equation-aware, transformer-based** summarization system tailored specifically for LaTeX scientific papers.

System Output Demonstration



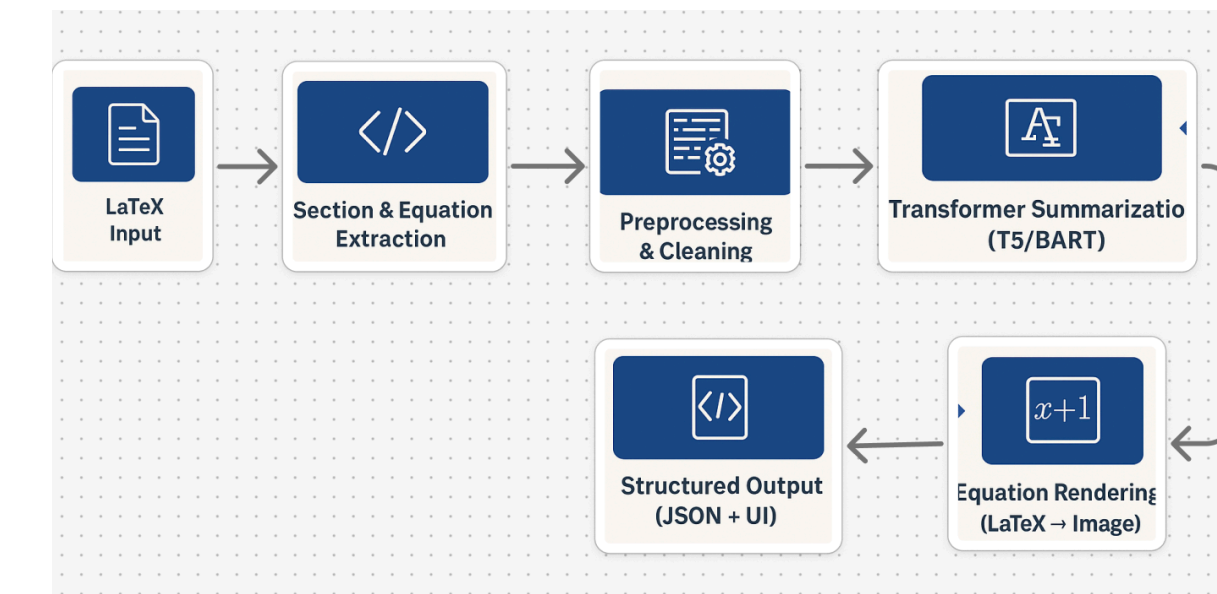
• Fig. 1

$$(x, y) \mapsto \left(x + \frac{y}{x}, x + y + \frac{2y}{x} + \frac{1}{x} + \frac{1}{y}\right)$$

• Fig. 3

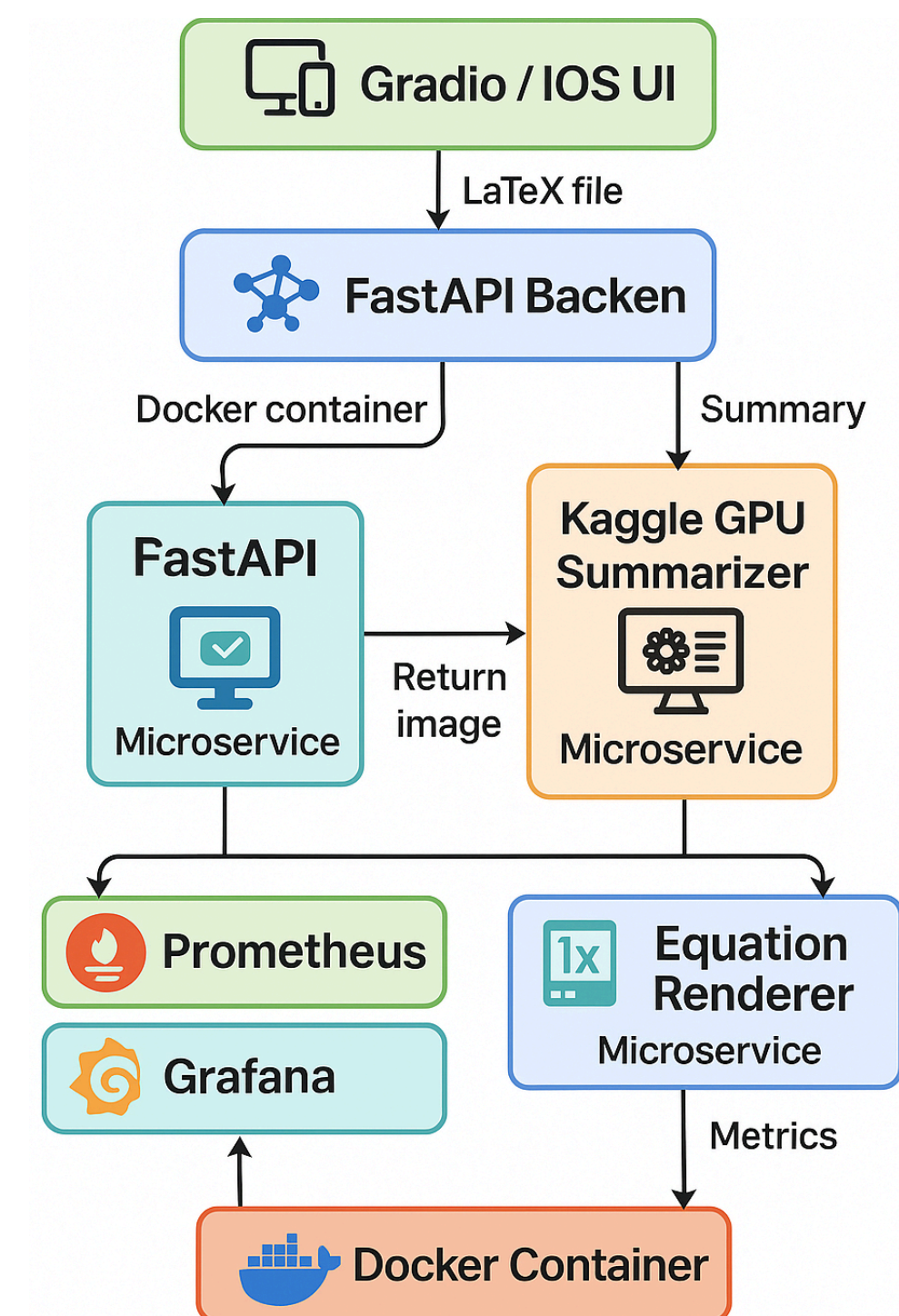
- Fig. 1. Gradio User Interface**
Shows the interactive UI where users upload LaTeX files and view generated summaries.
- Fig. 2. Section-Level Summary Output**
Displays the transformer-generated scientific summaries produced by the T5/BART model.
- Fig. 3. Rendered Equation Example**
Demonstrates how LaTeX equations are converted into clean PNG images for improved readability.

Model Pipeline



- Structured Output Generation**
Summaries and metadata are packaged into a structured JSON response.
- LaTeX Extraction & Preprocessing**
Sections, subsections, and mathematical equations are extracted and cleaned from the .tex document.
- Transformer-Based Summarization**
Clean scientific text is summarized using T5/BART to generate concise, section-level summaries.
- LaTeX Extraction & Preprocessing**
Important LaTeX equations are converted into readable images and displayed in the Gradio/iOS interface.

Architectural Diagram / Workflow



- FastAPI Backend acts as the central controller**, receiving LaTeX input, extracting sections and equations, performing preprocessing, and coordinating communication between the summarizer and renderer microservices.
- A remote Kaggle GPU Summarization Service** executes the heavy transformer models (T5/BART), enabling fast, GPU-accelerated generation of scientific summaries without requiring local high-end hardware.
- The Equation Renderer Microservice** processes mathematical expressions extracted from LaTeX and converts them into clean, high-resolution PNG/SVG images, making complex equations easier to read and interpret.
- All system components run inside Docker containers**, providing isolation, consistency, reproducibility, and simplified multi-service deployment across different environments.
- Prometheus and Grafana form the monitoring layer**, capturing API latency, request volume, and custom metrics like `summariq_feedback_total` to support real-time performance visualization and system health tracking.

Implementation Results

- T5-small and BART-large models** successfully generated concise section-level summaries with consistent scientific coherence.
- Equation Renderer** produced high-quality PNG equation images for all extracted math expressions, improving readability for non-LaTeX users.

Metric	Value
ROUGE-1	39.02
ROUGE-2	15.72
ROUGE-L	24.15
Latency	< 5 sec
Equation Success	90%

GitHub Repository



LinkedIn



Github



Kaggle

